

CBSE Test Paper 04
Chapter 5 Laws of Motion

1. A man of mass 70 kg stands on a weighing scale in a lift which is moving upwards with a uniform acceleration of 5 m s^{-2} what would be the reading on the scale?**1**
 - a. 130 kg
 - b. 120 kg
 - c. 150 kg
 - d. 105 kg
 2. Impulse is **1**
 - a. the derivative of force over a short period
 - b. the integral of force over a short period
 - c. the difference of force over a short period
 - d. the average of force over a short period
 3. A rocket with a lift-off mass 20,000 kg is blasted upwards with an initial acceleration of 5.0 m s^{-2} . Calculate the initial thrust (force) of the blast. **1**
 - a. 300000 N
 - b. 350000 N
 - c. 378000 N
 - d. 365000 N
 4. A monkey of mass 40 kg climbs on a rope which can stand a maximum tension of 600 N. What is the tension in the rope if the monkey climbs up with an acceleration of 6 m s^{-2} **1**
 - a. 740 N
 - b. 760 N
 - c. 600 N
 - d. 640 N
 5. Friction is **1**
 - a. perpendicular to contact surface and aids motion
 - b. perpendicular to contact surface and opposes i motion
 - c. parallel to contact surface and opposes the relative motion
 - d. parallel to contact surface and aids motion
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6. A force of 36 dyne is inclined to the horizontal at an angle of 60° . Find the acceleration it will produce in a mass of 18 g that moves in a horizontal direction. **1**
7. Action and reaction forces do not balance each other. Why? **1**
8. Why does a child feel more pain when she/he falls down on a hard cement floor than when she/he falls on the soft muddy ground in the garden? **1**
9. Define impulse and impulse-momentum theorem. **2**
10. A body of mass 2 kg is being dragged with a uniform velocity of 2 ms^{-1} on a rough horizontal plane. The coefficient of friction between the body and the surface is 0.2. Calculate the amount of heat generated per second. Take $g = 9.8 \text{ ms}^{-2}$ and $J = 4.2 \text{ Jcal}^{-1}$. **2**
11. A girl riding a bicycle along a straight road at a speed of 5 m s^{-1} throws a stone of mass 0.5 kg which has a speed of 15 ms^{-1} with respect to the ground along with her direction of motion. The mass of the girl and bicycle is 50 kg. Does the speed of the bicycle change after the stone is thrown? What is the change in speed, if so? **2**
12. Show that Newton's third law of motion is contained in the second law. **3**
13. An aircraft executes a horizontal loop at a speed of 720 km/h with its wings banked at 15° . What is the radius of the loop? **3**
14. Obtain an expression for the centripetal force required to make a body of mass m moving with a speed v around a circular path of radius r . **3**
15. Figure shows the position-time graph of a body of mass 0.04 kg. Suggest a suitable physical context for this motion. What is the time between two consecutive impulses received by the body? What is the magnitude of each impulse? **5**

